Math 125 **Midterm 2** **Burdzy, Autumn 2022**

Your Name

Your Signature

Student number

- Turn off all cell phones, pagers, radios, mp3 players, and other similar devices.
- This exam is closed books. You may use one $8\frac{1}{2}" \times 11"$ sheet of handwritten notes (you can write on both sides).
- You can use only Texas Instruments TI-30X calculator.
- Give your answers in exact form, not decimals, unless instructed otherwise.
- In order to receive credit, you must **show all of your work**. If you do not indicate the way in which you solved a problem, you may get little or no credit for it, even if your answer is correct.
- **Check your work carefully.** We will award only limited partial credit.
- Place $\boxed{\text{a box around your answer}}$ to each question.
- If you need more room, use the backs of the pages and indicate that you have done so.
- Raise your hand if you have a question.
- This exam has 6 pages, plus this cover sheet. Make sure that your exam is complete.

1. (10 points) Compute the integral.

$$\int (4x^2 - 9)^{-5/2} dx$$

2. (10 points) Compute the integral.

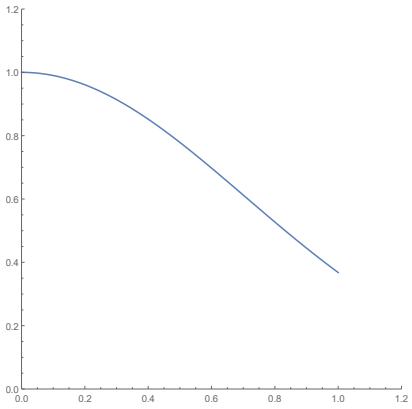
$$\int \frac{x^{20}+x^{18}+1}{x^3+x}\,dx$$

3. (10 points) In a physics experiment, a non-moving particle was isolated. Then it was given the acceleration of $\sin^2 t$ m/sec^2 for $\pi/2$ seconds, where t is the time in seconds after the start of the motion. Find the average velocity of the particle during the first $\pi/2$ seconds.

   Give your answer in the exact form or as a decimal number with 5 significant digits.

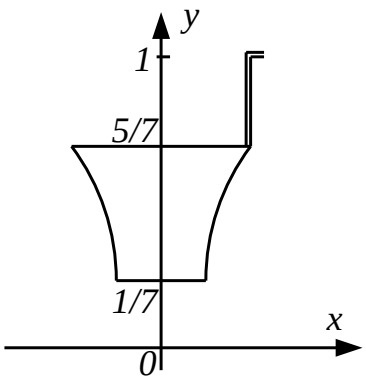
4. (10 points) The region A is below the curve $y = e^{-x^2}$, above the x-axis, between $x = 0$ and $x = 1$. Write an integral representing the volume of the body of revolution obtained by rotating the region about the x-axis. Apply the disc/washer/annulus method. Do not evaluate the integral. Instead, estimate the volume using Simpson's method with 4 subintervals.

   Give your answer as a decimal number with 5 significant digits.



5. A region D lies above $y = 1/7$, below $y = 5/7$, to the right of the y-axis, and to the left of the graph of $x = \dfrac{1}{\sqrt{y}(1-y)}$. A vessel is a body of revolution obtained by rotating the region D about the y-axis. The vessel is filled with water. The water needs to be pumped through the spout located 1 meter above the ground. Note that the bottom of the vessel is $1/7$ meters above the ground.

   The dimensions are in meters. Recall that the mass density of water is 1,000 kg/m^3 and the gravitational acceleration is 9.8 m/s^2.



   (a) (4 points) How much work is needed to pump out a single layer of water y meters above the ground, with thickness Δy, through the spout?

   (b) (2 points) Write a formula for the work needed to pump all the water through the spout.

   (c) (4 points) Find the total work by evaluating the integral. Give the answer in the exact form or in the decimal form with 5 significant digits.

6. (10 points) A region A lies above the x-axis, below the graph of $y = \cos^2 x$, between $x = \pi/4$ and $x = \pi/2$ (see the figure below). Find the volume of the solid of revolution obtained by rotating the region A about the y-axis.

   Give the answer in the exact form or in the decimal form with at least 5 significant digits.

